Master of Computer Applications

Semester – III

Data Analytics for Business

Final Project

Title: Creating a Used Car Sales Analytics Dashboard

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Used Car Sales Analytics Dashboard

Developing the Used Car Sales Analytics Dashboard

Problem Statement and Data Collection:

Given Problem Statement:

You are a Business Intelligence Consultant for a growing organization. Your task is to develop an end-to-end Business Analytics solution using a dataset of your choice to support data-driven business decisions.

Project-Specific Problem Statement:

A used-car dealership's inventory and transaction data is spread across raw sales records with no consolidated view for management, making it difficult to track which cars are selling, which are sitting unsold, and how profitability varies across manufacturers, car types, energy types, locations, and sales agents. The task is to import this data into Power BI, clean and model it, and build an interactive dashboard that shows total sales revenue, profit, sold/unsold inventory, and agent- and manufacturer-wise performance to support quick, data-driven decisions.

Data Description:

Dataset used: Used Car Sales data, loaded into Power BI and modelled as the used_car_sales_db table (used_car_sales_db used_car_sales). Each row represents one car listed for sale by the dealership. Key columns include:

- Sale_Status — whether the car has been “Sold” or is still “Unsold” in inventory.
- sold_price and purchased_price — the price the car was sold for and the price the dealership originally paid, used to compute revenue, cost and profit.
- manufacturer — the brand of the car (e.g., Maruti, Toyota, Hyundai).
- car_type — the body style/segment of the car (Convertible, Hatchback, Sedan, SUV, Truck).
- energy_type — the powertrain of the car (Petrol, Diesel, Electric, Hybrid).
- mileage_km — the total mileage of the car in kilometres.
- sales_agent — the dealership representative who handled the sale.
- location — the geographic location tied to each sale, used for the location map.
- model_year / Year — the year associated with each sale, used to build the yearly revenue trend.

Data Cleaning & Transformation (Power Query):

The used car sales data was connected in Power BI and loaded into the used_car_sales_db used_car_sales table. The following transformations were applied in Power Query Editor before loading to the model:

- Previewed the table to confirm the columns — Sale_Status, sold_price, purchased_price, manufacturer, car_type, energy_type, mileage_km, sales_agent and location.
- Promoted Headers — the first row of the incoming data was promoted to become column headers.
- Changed Type — each column's data type was set correctly (text for manufacturer, car_type, energy_type, Sale_Status and sales_agent; decimal/whole number for sold_price, purchased_price and mileage_km).
- Standardised the Sale_Status column to exactly two values, “Sold” and “Unsold”, so the sold/unsold KPI cards and the Total Cars donut chart calculate correctly.
- Verified there were no fully-blank rows and no duplicate records before closing & loading to the model.

Data Modelling:

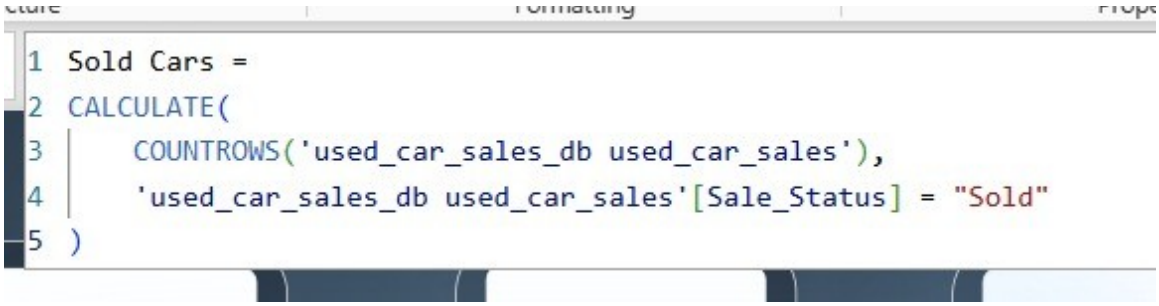
The model uses a single fact table built directly from the cleaned source:

- used_car_sales_db used_car_sales (fact table) — one row per car, holding Sale_Status, sold_price, purchased_price, manufacturer, car_type, energy_type, mileage_km, sales_agent and location.
- No additional dimension tables were required — the manufacturer and car_type slicers, along with all visuals, filter directly off this single table, keeping the model simple and performant.

DAX Measures and KPIs:

1. Sold Cars:

Counts the number of cars whose Sale_Status is “Sold”.

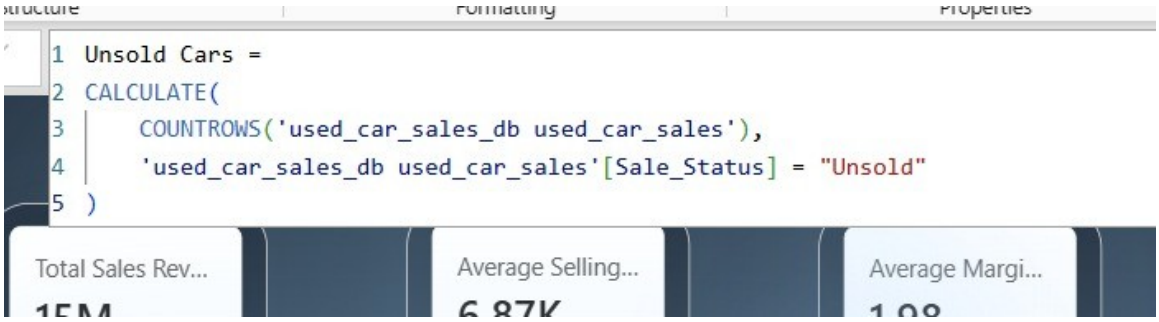


Screenshot Image: Sold Cars DAX measure

```
Sold Cars =
CALCULATE(
    COUNTROWS('used_car_sales_db used_car_sales'),
    'used_car_sales_db used_car_sales'[Sale_Status] = "Sold"
)
```

2. Unsold Cars:

Counts the number of cars still sitting in inventory with a Sale_Status of “Unsold”.

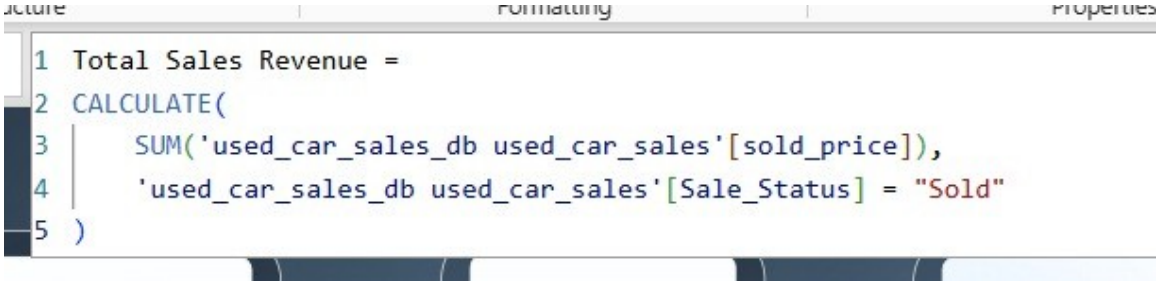


Screenshot Image: Unsold Cars DAX measure

```
Unsold Cars =
CALCULATE(
    COUNTROWS('used_car_sales_db used_car_sales'),
    'used_car_sales_db used_car_sales'[Sale_Status] = "Unsold"
)
```

3. Total Sales Revenue:

Sums the sold_price of every car that has been sold, giving the dealership's realised revenue.

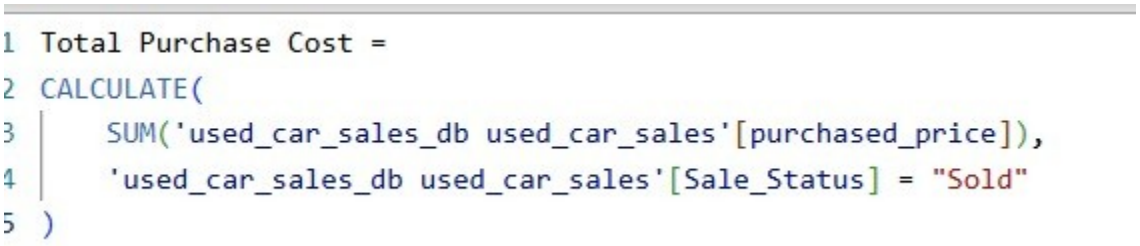


Screenshot Image: Total Sales Revenue DAX measure

```
Total Sales Revenue =
CALCULATE(
    SUM('used_car_sales_db used_car_sales'[sold_price]),
    'used_car_sales_db used_car_sales'[Sale_Status] = "Sold"
)
```

4. Total Purchase Cost:

Sums the purchased_price of every car that has been sold, giving the dealership's total acquisition cost for sold inventory.



Screenshot Image: Total Purchase Cost DAX measure

```
Total Purchase Cost =
CALCULATE(
    SUM('used_car_sales_db used_car_sales'[purchased_price]),
    'used_car_sales_db used_car_sales'[Sale_Status] = "Sold"
)
```

5. Total Profit:

Derived as the difference between realised revenue and the corresponding purchase cost, and used to power the Total Profit KPI card.

```
Total Profit =
[Total Sales Revenue] - [Total Purchase Cost]
```

6. Average Selling Price, Average Margin & Sales Conversion:

Three supporting measures feed the top KPI strip of the Sales & Revenue Analysis page:

- Average Selling Price = AVERAGE of sold_price across Sold cars — 6.87K.
- Average Margin = average ratio of (sold_price − purchased_price) to purchased_price across Sold cars — 1.98.

- Sales Conversion = DIVIDE([Sold Cars], [Total Cars]) — the share of listed cars that convert to a sale, currently 0.22 (22%).

Together these measures drive the KPI cards — Total Cars, Sold Cars, Unsold Cars, Total Sales Revenue, Total Purchase Cost and Total Profit — seen on the dashboard below.

Methodology:

All KPI cards, charts, and slicers were arranged into two connected interactive pages of the Used Car Sales Analytics dashboard in Power BI:

Page 1 — Used Car Sales Analytics Dashboard (inventory view):

- Total Cars, Sold Cars, Unsold Cars, Total Sales Revenue and Total Profit KPI cards.
- Total Sales Revenue by Year (Line Chart).
- Total Cars by Sale_Status (Donut Chart, Sold vs Unsold).
- Sold Cars by car_type (Bar Chart).
- Total Sales Revenue by manufacturer (Bar Chart).
- Total Sales Revenue by sales_agent (Bar Chart).
- Sold Cars by location and Sale_Status (Map).
- Manufacturer and car_type slicers/filters on the left-hand filter panel.

Page 2 — Sales & Revenue Analysis (portfolio view):

- Total Sales Revenue, Average Selling Price, Average Margin and Sales Conversion KPI cards.
- Total Sales Revenue by Year (Line Chart, 2016–2024).
- Total Sales Revenue by manufacturer (Bar Chart).
- Sum of mileage_km, Sum of price and Sold Cars by car_type (Scatter Chart).
- Sold Cars by energy_type and car_type (Stacked Column Chart).

The dashboard enables management to analyse revenue and inventory health across manufacturers, car types, energy types and locations, monitor sold-vs-unsold stock and profitability, and identify top-performing sales agents, supporting data-driven decisions for the dealership.

Interactive Dashboard Images:

- The Used Car Sales Analytics dashboard provides an interactive overview of inventory and sales-agent performance, filterable by manufacturer and car type.
- Displays key metrics such as Total Cars, Sold Cars, Unsold Cars, Total Sales Revenue and Total Profit.
- Shows how Total Cars split between Sold and Unsold status via the donut chart.
- Ranks sales agents and manufacturers by Total Sales Revenue, and plots Sold Cars by location on a map.



Screenshot Image: Used Car Sales Analytics Dashboard

- The Sales & Revenue Analysis page gives a portfolio-level view across the full manufacturer base.
- Tracks Total Sales Revenue by Year from 2016 to 2024, alongside Average Selling Price, Average Margin and Sales Conversion.
- Compares Total Sales Revenue by manufacturer, and breaks down Sold Cars by energy_type and car_type.



Screenshot Image: Sales & Revenue Analysis

Power BI Service Workspace:

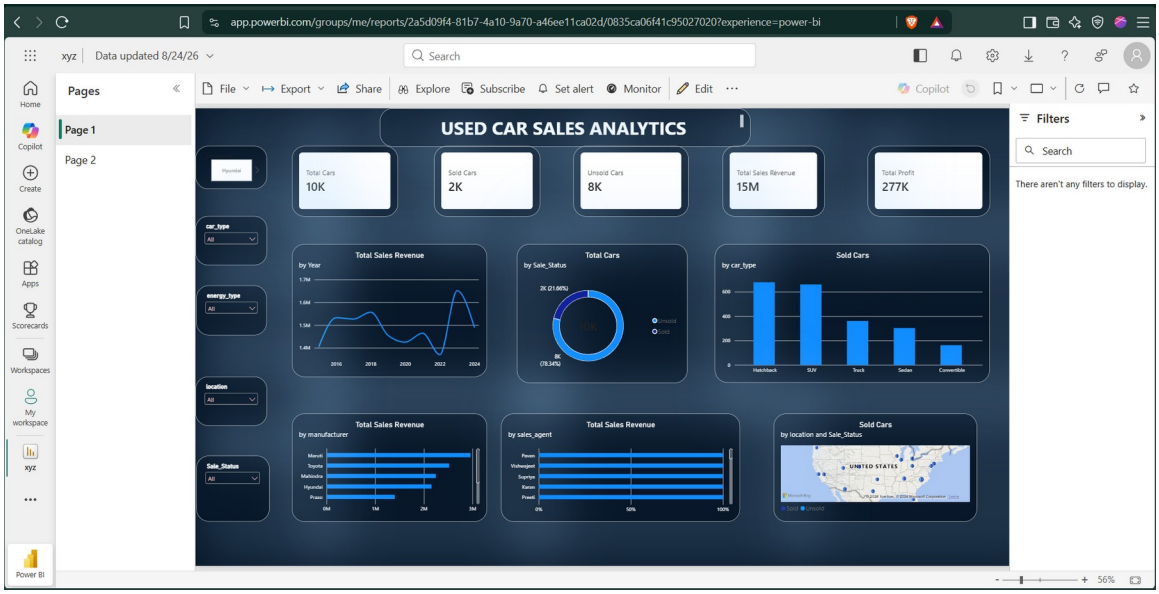
- Published the Used Car Sales Analytics dashboard to the "xyz" workspace in Power BI Service for online access and sharing.

- The dashboard is available in the workspace with interactive KPIs, charts, slicers and filters, matching the desktop version across both pages.

The published report can be accessed through the following Power BI Service link:

[View Used Car Sales Analytics Dashboard in Power BI Service]

- The Power BI Service enables the dashboard to be viewed, shared, and accessed online from the workspace.



Screenshot Image: Used Car Sales Analytics Dashboard — Power BI Service

Business Question Answered:

I. What is the overall sales and profitability performance across the dealership?

Answer: Across the full portfolio, the dealership generated $15M in Total Sales Revenue, with an Average Selling Price of $6.87K per car, an Average Margin of 1.98, and a Sales Conversion rate of 0.22 (22% of listed cars convert to a sale) — indicating healthy revenue but room to improve conversion of listed inventory.

II. Which manufacturer contributes the most to total sales revenue?

Answer: Maruti is the strongest manufacturer at roughly $2.8M in Total Sales Revenue, ahead of Toyota at roughly $2.2M, making Maruti the clear revenue leader in the current manufacturer mix.

III. How does the yearly sales revenue trend look, and when did it peak?

Answer: Total Sales Revenue by Year fluctuates between roughly $1.4M and $1.6M from 2016 through 2022, then rises sharply to a peak around 2023 before easing slightly into 2024 — pointing to a recent, strong uptick in yearly performance.

IV. How is sold inventory distributed across energy type and car type?

Answer: Sold Cars are fairly evenly spread across Electric, Petrol, Hybrid and Diesel energy types, each contributing roughly 500 sold units, with Convertible, Hatchback, Sedan, SUV and Truck body types each present within every energy type — showing a balanced, diversified sales mix rather than dependence on a single powertrain or segment.

V. What does the sold-vs-unsold inventory split look like, and is stock moving efficiently?

Answer: On the inventory dashboard (filtered to Hyundai), Total Cars stands at 2K against only 358 Sold Cars and roughly 1K Unsold Cars — a sold share of about 24% — signalling a meaningful backlog of unsold stock that ties up capital and warrants closer review of pricing or marketing for slow-moving models.

VI. Which sales agents are driving the most revenue, and which car type sells best?

Answer: Hariom leads Total Sales Revenue by sales_agent, followed by Sai, Pavan, Bhavani and Anushka in that order. At the manufacturer/car-type level shown for Hyundai, Hatchback is by far the best-selling body style, indicating agent performance and body-style demand are both worth tracking together when planning inventory and incentive programs.

Business Insights and Recommendation:

Key Insights:

- Total Sales Revenue is heavily concentrated in a small number of manufacturers — Maruti and Toyota together dominate the portfolio, while a low overall Sales Conversion rate (0.22) suggests a large share of listed cars are not converting into sales.
- The sharp jump in yearly revenue around 2023 stands out against a relatively flat 2016–2022 trend, suggesting a specific pricing, marketing, or inventory strategy from that period that may be worth replicating.
- Sold Cars are well diversified across energy types (Electric, Petrol, Hybrid, Diesel) and body types, showing the dealership is not overexposed to any single segment of the market.
- A large unsold inventory relative to sold units (roughly 1K unsold vs 358 sold in the filtered view) indicates capital tied up in slow-moving stock that is not being addressed quickly enough.
- Performance varies meaningfully across sales agents, with Hariom significantly outpacing the lowest-ranked agents, pointing to a gap in either sales technique, territory, or lead quality that could be closed with coaching or reallocation.

Recommendations:

- Investigate the pricing, marketing, or sourcing changes behind the 2023 revenue spike and evaluate whether the same approach can be extended into 2024 and beyond.
- Set a target to raise the Sales Conversion rate above the current 0.22 by reviewing pricing on slow-moving Unsold inventory and prioritising promotion of aged stock.
- Use the Total Cars by Sale_Status donut chart as a recurring health check — flag any manufacturer or car_type where the unsold share grows materially, and act before it becomes a large capital drag.
- Study what makes Maruti and top agents like Hariom successful, and apply those playbooks (stocking mix, agent training, customer outreach) to underperforming manufacturers and agents.
- Continue monitoring energy_type and car_type mix so the dealership keeps its currently balanced inventory profile as demand shifts toward electric and hybrid vehicles.

Conclusion:

This project delivers an end-to-end Business Intelligence solution for a used-car dealership's sales and inventory operations: transaction-level data was cleaned and typed in Power Query, and modelled into a single used_car_sales_db used_car_sales table enriched with DAX measures for Sold Cars, Unsold Cars, Total Sales Revenue, Total Purchase Cost and Total Profit. The resulting two-page interactive Power BI dashboard lets management move from a portfolio-level view of $15M in Total Sales Revenue, Average Selling Price and Sales Conversion, down to a single manufacturer, car

type, or sales agent in a few clicks — surfacing insights that can directly inform inventory pricing, sales-agent coaching, and manufacturer stocking strategy.

Project Links:

- **Power BI Dashboard (live report):** https://app.powerbi.com/links/zErL1PlIO9?ctid=6bab7765-c387-4d0d-9be7-7f5679251594&pbi_source=linkShare
- **GitHub Repository (project files):** https://github.com/Amina872/Power-BI/tree/main/used-car-sales-analytics-powerbi-final-project